

Fall 2020
THE UNIVERSITY OF TULSA
MATH 7013/5013, ADVANCED MATHEMATICAL MODELING

INSTRUCTOR: Dr. Dale R. Doty
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PHONE: ***-***
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CONTACT OPTIONS: Harvey, Collaborate and Email.

OFFICE HOURS: Collaborate in Harvey: TBA, CDT.
Any time by email (response within 24 hours).

CLASS HOURS: Online asynchronously; study the lessons and videos and do the homework on your own schedule and at your own pace, but stay current.

REQUIRED TECHNOLOGY:

Accessing Course Content: Access to the Internet (Personal Computer, Laptop, Tablet, Cell Phone or Streaming TV etc.)

Returning Examinations: Printer/Scanner or Cell Phone and Email.

COLLABORATE (within Harvey): Follow these steps:

Within your (Chrome) browser enter: **Harvey.utulsa.edu**

Login with your TU Username and Password.

Within Harvey select **My Courses**.

Select **Math-7013-01 (20/FA)**.

Click: The **arrow**, upper left, below the blue line hamburger.

Scroll down to **COLLABORATE** and click.

Select within Sessions: **Math-7013-01 (20/FA)**. Make sure the **Course Room** is unlocked (open only during office hours) so that you can enter.

Join Course Room: Click.

Microphone and Camera; On/off at the bottom.

EXAMINATIONS: No examinations will be given, the course grade will be determined from the homework and semester project grades.

HOMEWORK: All material covered in the course will have assigned homework. All homework will be submitted as an email attachment to Dale-Doty@utulsa.edu. (Not returned in Harvey).

HOMEWORK DUE DATE: Each homework due date will be posted in the syllabus schedule.

LATE HOMEWORK: Late homework will be penalized 20% for each class day it is late.

SEMESTER PROJECT: Each student will define, analyze, develop and implement a semester long project. The completed project is to be submitted to the class and evaluated by the instructor.

TEXTS

Mathematics for Dynamic Modeling, by Beltrami; Academic Press, Inc.
ISBN: 9780120855667

COURSE LEARNING OUTCOMES:

Upon completion of the course, a student should be able to:

Understand and utilize the following basic programming paradigms: List, Functional, Recursive, Pattern Matching and Arbitrary Precision programming.

Understand and utilize the following intermediate programming paradigms: Scientific/Technical Graphics and Computational Algorithmic Efficiency.

Understand and utilize the following advanced programming paradigms: Scientific/Technical Algorithms, Scientific/Technical Presentations and Real-Time Computational Dynamics.

Understand and be able to implement fundamental modeling techniques.

Be able to transform a model into a computational environment for the purposes of understanding and analysis.

Develop the skills necessary to generate a Scientific/Technical Presentation.

Mathematical Modeling: A traditional math course starts with a predefined mathematical problem (e.g. those found in Ordinary Differential Equations and Partial Differential Equations etc. without much motivation as to their origin) and illustrates how these problems can be solved by teaching solution procedures. Unlike a traditional math course, this course starts with the problem and illustrates how it can be transformed into a mathematical model. Knowledge of solution procedures is not required for this course as the emphasis is on the modeling process and not on the solution procedure. Solutions will be found automatically within the context of Scientific/Technical programming.

Scientific/Technical Presentations: Mathematical modeling must ultimately include the ability to create a Scientific/Technical Document and Presentation. This is similar in concept to TeX and PowerPoint, but is specifically designed to incorporate numerical, technical, graphical, computational and dynamic information that is not naturally part of these limited environments. The course will conclude with a semester-long project (a topic specifically selected by the student to support their educational and personal interests) ultimately generating a report as well as a presentation.

Background and Prerequisites: The course is designed to be accessible to all students from upper level undergraduate through the graduate level. Upper level math courses (including Linear Algebra) are not explicitly required as the emphasis is on modeling and not on mathematical procedures. However, as a prerequisite a student should have completed courses in the Calculus. If further clarification on background, prerequisites or content is required, please contact Dr. Doty (dale-doty@utulsa.edu).

COURSE GRADE:

The course grade will be the accumulated average of the following two items: Equally weighted Homework Assignments and the Semester Project. The semester project will have a weighting equivalent of four homework assignments.

CLASS ATTENDANCE:

Attendance will not be taken as the course is online and asynchronously.

MISSED WORK:

If a student is absent or unavailable for required work and has a valid excuse (e.g., an excuse acceptable to the Center for Student Academic Support), then an arrangement to make up the work can be made.

ACADEMIC MISCONDUCT:

The College of Engineering and Natural Sciences policy on academic misconduct will be followed. <https://portal.utulsa.edu/colleges/engineering-and-natural-sciences/layouts/15/start.aspx#/SitePages/ENS%20Undergraduate%20Academic%20Misconduct%20Policy.aspx>

STUDENT CENTER FOR ACADEMIC SUPPORT:

Students with disabilities should contact the Center for Student Academic Support to self-identify their needs in order to facilitate their rights under the Americans with Disabilities Act.

All students are encouraged to familiarize themselves with and take advantage of services provided by the Center for Student Academic Support such as tutoring, academic counseling, and developing study skills. The Center for Student Academic Support provides confidential consultations to any student with academic concerns as well as to students with disabilities.

Advanced Mathematical Modeling - Math 7013/5013

Fall 2020 Tentative Schedule

<u>Week</u>		<u>Monday</u>	<u>Tuesday</u>	<u>Wednesday</u>	<u>Thursday</u>	<u>Friday</u>
August	24	Lesson 00		Lesson 01 Due: L00		Lesson 02 Due: L01
	31	Lesson 03 Due: L02		Lesson 04 Due: L03		Lesson 05 Due: L04
September	7	Holiday		Lesson 06 Due: L05		Lesson 07 Due: L06
	14	Lesson 08 Due: L07		Lesson 09 Due: L08		Lesson 10 Due: L09
	21	Lesson 11 Due: L10		Lesson 12 Due: 11		Lesson 13 Due: L12
	28	Lesson 14 Due: L13		Lesson 15,16 Due: L14		Lesson 17 Due: L15,16
October	5	Book 1.1 Due: L17		Book 1.2		Book 1.3
	12	Book 2.1		Book 2.2 Due: Book 1.4		Book 2.3
	19	Book 3.1		Book 3.2 Due: Book 2.4		Book 3.3
	26	Book 3.4		Book 3.5 D		Book 5.1
November	2	Due: Book 3.5E		Book 5.2		Book 5.3
	9	Book 5.4		Book 5.6		Book 6.1
	16	Due: Book 5.8		Book 6.3		Due: Book 6.5
	23	Thanksgiving		Vacation		Week
	30	Book 7.1		Book 7.2		Book 8.1
December	7	Due: Book 7.5	No Final Exam	No Final Exam	Project Presentations	No Final Exam
	14	No Final Exam	No Final Exam	No Final Exam	No Final Exam	No Final Exam

KNOW YOUR TITIE IX

Sexual misconduct is prohibited by Title IX of the Educational Amendments of 1972 ("Title IX") and will not be tolerated within the TU community. Sexual misconduct encompasses all forms of sex and gender-based discrimination, harassment, violence, and assault, as well as dating violence, domestic violence, interpersonal violence, stalking and sexual exploitation. If you or someone you know has been harassed, assaulted, or stalked, or if you have questions about violence prevention resources available to you, please contact any of the following campus and community resources:

Title IX Coordinator 918-631-4602

Office of Violence Prevention 918-631-2324

TU Counseling and Psychological Services 918- 631-2241

Campus Security 918-631-5555

Domestic Violence Intervention Services 918-585-3163 or 918-743-5763

Tulsa Police Department 918-596-9222 or 911 (emergency situations)

For more information about your rights under Title IX, please visit our *Policies and Laws* page <https://utulsa.edu/sexual-violence-prevention-education/policies-laws/> on the TU website or contact the Title IX Coordinator. ***Every student on our campus has the right to resources.*** Please come forward and ask questions, report, and help us eradicate sexual misconduct and interpersonal violence by stopping the silence surrounding it.